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Abstract

Key words : *Ektara, Ravanhatta, Invisible Laboratory, Baul tradition, Bhopa tradition, Pabuji epic, Phad, Folk string instruments, Bowed instruments, Oral Tradition, Bejara, Gaj, Tata Vadya, Savoir-faire, Tumba, Lau, Guru-shishya parampara, Gharana.*

Many studies of science and technology in medieval India have focused primarily on textual and elite knowledge systems, often overlooking the material practices and artisan communities that sustained technological knowledge. While historiography offers only fragmentary references to musical instruments, it rarely addresses the processes of their making or the knowledge embedded within craft traditions. Drawing on ethnomusicological and material culture approaches, this study examines musical instrument-making as a site of scientific knowledge production in medieval India. Building on frameworks that recognise objects as carriers of cultural and technological meaning, this paper argues that artisans embedded empirical understandings of materials and acoustics within oral and hereditary traditions. In this study, I foreground the agency of artisan communities to better understand how knowledge of material science was constructed, transmitted, and adapted across regions. These knowledge systems were dynamic, shaped by ecological conditions and sustained through oral and performative modes of transmission.

Focusing on regional stringed instruments (*Tatā Vadya),* such as the Ravanhatta and Ektara, alongside the communities that produce and perform them, I explore how environmental factors and resource availability, shape their construction, structural design, and acoustic properties. This analysis considers regional variation, systems of knowledge transmission, and evolving perceptions of craftsmanship, while highlighting how material constraints generate specific sonic possibilities. By positioning craft practices and oral traditions as forms of scientific knowledge, this study challenges text-centric narratives of medieval Indian science and technology and demonstrates that musical instruments serve as key sites for understanding a localized, practice-based history of science.

Material Science, Craftsmanship, and Oral Knowledge Systems:

Musical Instrument-Making in Medieval India

Introduction

Histories of science and technology in medieval India have long been dominated by images of large-scale state projects. Scholarly attention has consistently gravitated toward monumental irrigation systems, fortified architecture, and the sophisticated machinery of warfare. These domains leave behind tangible archaeological remains and are extensively documented in royal chronicles like the *Akbarnama* and technical treatises such as the *Shukraniti*. These large-scale enterprises, often patronised by states and empires, have come to define what counts as "science"[1] and "technology"[2] in the premodern Indian context. The grand tanks of the Vijaynagara Empire or the metallurgical expertise evident in imperial artillery are thus regarded as indicators of scientific advancement. In contrast, smaller, everyday objects rarely enter the narrative as sites of scientific knowledge. A musical instrument, for instance, is more often treated as an artistic or cultural artefact than as a technological one.

This imbalance raises a critical question: why is a machine of war readily recognised as technology, while a device that produces sound is not? Both are carefully engineered objects, that rely on precise manipulation of materials and produce controlled effects through the application of underlying principles. Yet, one is categorized as science and technology, while the other is placed in the category of art or culture. This distinction is not inherent but historiographical, it reflects the ways in which modern frameworks have privileged certain forms of knowledge over others.

A musical instrument, at first glance, may appear to be an object of aesthetic expression rather than scientific inquiry. However, a closer examination reveals that the production of sound from materials such as wood, bamboo, gourd, or metal is far from spontaneous. It requires a careful calibration of form, tension, density, and resonance. Instruments such as the Ektara, with its seemingly simple structure, and the more complex Ravanhatta demonstrate intricate connections between material selection, structural design, and acoustic output. In this sense, they are not merely cultural objects but technological devices, engineered to produce specific sonic effects. To recognise this is to shift our perspective. Musical instruments must be understood not only as cultural artefacts but also as technological devices, designed to manipulate natural properties in order to achieve specific sonic effects. This shift requires us to rethink the very definitions of science and technology in the medieval context. Instead of limiting science to formal theories and written

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[1] According to the Cambridge Dictionary, "*science*" refers to the careful study of the structure and behaviour of the physical world, especially by watching, measuring, and doing experiments.

[2] According to Encyclopaedia Britannica, "*technology*" refers to the application of scientific knowledge to the practical aims of human life.

texts, it should also include forms of "practical science"[3]. Similarly, technology should be understood not only as large machines but also as everyday objects that reflect practical scientific understanding.

It is within this redefined framework that the idea of the "Invisible Laboratory"[4] becomes central. Much like Steven Shapin's notion of the "invisible technician," which highlights the overlooked role of assistants and craftsmen in the production of scientific knowledge, the medieval workshop can be understood as a dispersed and often unrecorded site of experimentation. This perspective resonates with the work of Iwan Rhys Morus, who emphasises the crucial role of instrument-makers and artisans as active contributors to scientific knowledge, highlighting that their material practices and technical skills were central to the production of knowledge, rather than merely auxiliary to it.[5] Unlike the modern laboratory, which is institutionalised, documented, and visible, the medieval workshop functioned as a dispersed and often unrecorded site of experimentation. Here, artisans worked with materials, tested their properties, and refined their techniques over generations. Their knowledge was not codified in formal treatises but was transmitted through oral traditions, apprenticeship, and embodied learning.

At the same time, it is important to avoid framing this argument in absolute terms. It would be inaccurate to claim that musical knowledge was entirely absent from written traditions. Texts such as the Nāṭya Sastra and later medieval works on musical treatises, do engage with music in systematic ways, offering classifications of instruments and discussions of performance. However, these texts are primarily concerned with the theory of music, such as raga, tala, and aesthetics, they offer only limited engagement with the actual processes of instrument-making. The practical knowledge of material selection, construction, and maintenance remained largely outside textual discourse and was instead preserved within oral and artisanal traditions. This suggests not an absence of textual knowledge, but a division between elite theoretical discourse and practical craft knowledge, the latter remaining largely within the domain of artisans. As a result, the contributions of artisans, whose expertise was grounded in practice rather than writing, have been systematically overlooked.

Furthermore, the perception of music as a "luxury" associated with courtly life must also be reconsidered. Although music in medieval India is often characterised as a courtly luxury associated with elite patronage, this perspective reflects a narrow historiographical focus. While refined musical performances flourished in royal courts, musical practices were also deeply embedded in everyday social and devotional life. Instruments such as the Ektara, used by itinerant performers and devotional communities, demonstrate that the production and use of musical sound extended beyond elite spaces. Ektara players moved across regions carrying their instruments with them, using music as a means of livelihood, spiritual expression, and social communication. Over time, the instrument became inseparable from their daily lives, shaping both their identity and routine practices. In a similar manner, the Ravanhatta was integral to the lives of its traditional performers, particularly itinerant storytellers who used it to narrate oral epics and local legends. Their performances were not occasional or restricted events but formed part of a continuous cultural

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[3] The term "*practical science*" is used here to denote knowledge grounded in experience, repeated practice, and craft traditions.

[4] The term "invisible laboratory" is developed in this study to describe dispersed and non-institutional sites of knowledge production embedded within artisanal and oral practices. The concept is inspired by Steven Shapin's notion of the "invisible technician," which highlights the often-overlooked role of artisans in the making of scientific knowledge. See Shapin, Steven. "The Invisible Technician." *American Scientist*, vol. 77, no. 6, 1989, pp. 554–563.

[5] Morus, Iwan Rhys. "Invisible Technicians, Instrument Makers, and Artisans," in A Companion to the History of Science, ed. *Bernard Lightman. Oxford: Wiley-Blackwell,* 2016. Pp. 97-110.

practice embedded in everyday life, rituals, and community gatherings. Thus, both instruments illustrate how music functioned not merely as elite entertainment but as a lived, mobile, and socially embedded practice.

This wider social presence of music complicates the assumption that it was inherently elite or exclusive. Rather, its later association with prestige appears to be the result of its selective circulation within courtly and elite cultural frameworks. What came to be recognised as "high" art was therefore not defined by the nature of music itself, but by the social contexts in which certain forms were preserved, patronised, and recorded. At the same time, the material practices underlying instrument making remained widely distributed across artisan and performing communities. This reveals a more complex and inclusive landscape of knowledge production, in which technical skill and musical creativity were sustained beyond the boundaries of elite institutions.

This study, therefore, seeks to reposition musical instrument making as a cornerstone of medieval Indian science and technology. It argues that the crafting of instruments constituted a sophisticated form of material and acoustic science, expressed through technological design and sustained through oral and craft-based traditions. By focusing on the interplay between materials, sound, and practice, this section introduces the concept of Sound Technology as a legitimate domain of scientific inquiry, one that operates outside the boundaries of formal textual codification yet remains deeply systematic and empirically grounded. In doing so, the section lays the foundation for a broader re-evaluation of how knowledge is defined, located, and transmitted.

The Historiographical Problem: Textual Privilege and European Classifications

The way we write about science in India has been shaped for centuries by ways of thinking that come from outside India itself. When historians look at scientific knowledge, they tend to focus on what was written down in books, how knowledge was organised into neat categories, and how different types of learning were separated from each other. This approach has led to a particular problem: scientific knowledge gets sorted into fixed boxes like medicine, astronomy, mathematics, alchemy, physics, agriculture, botany, zoology, geography, gemology, architecture, encyclopaedia, and dictionary.[6] These categories might seem natural to us today, but they actually come from European ways of organising knowledge that developed after the Enlightenment period. When we apply these modern silos to medieval India, we are forcing a way of thinking onto a society that organised knowledge very differently.

This rigid way of classifying knowledge creates a serious distortion in how we understand intellectual life in medieval India. By forcing medieval inquiries into modern categories, we create artificial divisions that didn't exist for the people of that time. In reality, medieval Indian thinkers and craftspeople did not see their work as divided into separate, isolated fields. Instead, knowledge flowed freely between what we today would call different "disciplines." A single craftsman might be doing several things at once that we would now classify as completely separate sciences. Consider someone making a musical instrument: they need to understand which wood works best for sound (what we would call botany), how to manage the tension of strings and the resonance of the body (physics), and how to treat materials with oils and resins to preserve them and improve their sound (chemistry). To the medieval craftsman, these were not separate fields of study, they were simply different aspects of the same practical problem. But when historians try to fit this interconnected way of working into our modern categories, something important gets lost.

This problem becomes even clearer when we look at how historians have studied science and technology in medieval India. There has been a strong tendency to focus on certain areas while ignoring

[6] Trompf, G.W. "The classification of the sciences and the quest for interdisciplinarity: a brief history of ideas from ancient philosophy to contemporary environmental science." *Environmental Conservation,* Vol 38(2). 2011. pp.113-114.

others. The fields that get the most attention are agriculture, metallurgy, military technology, and shipbuilding.[7] These are treated as the main places where scientific innovation happened during this period. But why these particular fields? The answer tells us something important about how history gets written. These areas leave behind physical evidence that survives to the present day, we can still see irrigation canals, examine metal objects, study weapons, and look at ship designs. They also tend to be things that were commissioned or supported by rulers and governments, which means there are written records about them. And they are closely connected to state power and economic production, which makes them seem "important" in a way that other activities might not. For example, the Vijayanagara Empire built massive irrigation systems that transformed agriculture across southern India. These works were impressive engineering achievements, and they were documented because they served the empire's political and economic goals. Similarly, the Mughal Empire developed sophisticated metallurgy for making cannons and other weapons.[8] These technologies were “visible” not just because we can find physical examples of them today, but because they were central to imperial power and state formation. When historians focus on these areas, they are not wrong to do so, these were indeed important sites of scientific and technological knowledge. However, this focus on "state-sponsored science" narrows our understanding of what counted as science during this period.

Many other forms of knowledge were equally systematic and equally experimental, but they have been marginalized in historical writing because they do not fit the conventional model of what "science" should look like. These forms of knowledge were embedded in everyday life, in artisanal workshops, in performance traditions, and in practices that were passed down through experience rather than written textbooks. A potter developing new glazes through trial and error, a dancer understanding the physics of balance and movement, a weaver experimenting with dye combinations, a cook perfecting recipes through careful observation, these were all engaged in forms of inquiry that involved observation, experimentation, and the development of systematic knowledge. But because this knowledge was not written down in formal treatises, because it was not supported by state patronage, and because it does not fit neatly into modern categories of "science," it has often been overlooked or dismissed as merely "craft" or "tradition" rather than genuine scientific knowledge.

Much of the scientific knowledge in medieval India was never written down. Instead, it lived in the hands and minds of craftsmen. It was shared orally, through apprenticeships and through the direct transmission of practical skills from one generation to the next. A master craftsman would teach an apprentice not by giving them a book to read, but by showing them how to do things, correcting their mistakes, and helping them develop the feel and judgment that comes only from years of practice. Because this knowledge wasn't in books, it has often been dismissed as mere "craft" or "intuition" rather than "real" science. As a result, vast areas of scientific knowledge remain invisible to historians, not because they did not exist, but because the tools we use to look for them are not designed to find them.

To correct this, we can think of the artisan’s workshop as an "Invisible Laboratory." Their workshop was, in effect, a laboratory, though not the kind of laboratory that modern science recognises. In this space, the craftsman conducted systematic experiments every day. Whether it was a weaver figuring out the mathematics of a complex pattern or a potter experimenting with the chemical properties of a new glaze, these individuals were engaged in a form of inquiry that was both experimental and rigorous . They developed an "experiential truth" that was passed down for generations, creating a tradition of "Practical Science" that operated alongside formal academic learning. When we take this broader view, the history of science in India begins to look very different. We see a world where knowledge was not divided into rigid

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[7] Kuppuram, G., and K. Kumudamani, editors. “History of Science and Technology in India.,” Vol. III. *Sundeep Prakashan, Delhi,* 1990. Introduction.

[8] Datta, Pankaj Kumar. “Technological Aspects of Some Firearms in Mughal India.” *History of Science and Technology in India*, edited by G. Kuppuram and K. Kumudamani, Vol. III. *Sundeep Prakashan, Delhi,* 1990. pp. 175-184.

categories but flowed freely between different domains of practice. We see scientific inquiry happening not just in royal courts, but also in the workshops of craftspeople and the everyday lives of ordinary people. We see innovation driven not just by the demands of state power but also by the practical needs of communities and the curiosity of individuals.

This “Invisible Laboratory” challenges the assumption that the absence of detailed written manuals indicates the absence of scientific reasoning. Several important musical treatises from the medieval period—such as *Sangita Makaranda* attributed to Narada (7th -9th century CE), Nanyadeva’s *Bharata Bhashyam* (11th-12th century CE), *Matanga’s Brihaddeshi* (6th–8th century CE), *Jayadeva’s Gita Govinda* (12th-century CE), and Sarangadeva’s *Sangita Ratnakara* (13th century CE), rarely discuss the making of musical instruments or the material science practiced by artisanal communities. These texts were largely produced in courtly and elite contexts, so they reflect the perspectives of educated, literate groups while overlooking the practical and technical knowledge present within local communities. The gap between textual knowledge and craft practice suggests that a significant portion of technical expertise existed outside the domain of formal writing traditions. Instead, it was preserved and transmitted through oral traditions, apprenticeship systems, and embodied learning, that leave fewer archival traces.

Musical instruments provide one of the best examples of this "Practical Science.” The crafting of musical instruments in medieval India was not merely an act of artistic intuition or cultural expression, it required a precise and experience-based understanding of sound that we would today describe as acoustics.[9] Instrument makers had to understand how different materials affected sound quality, how the shape and size of an instrument's body influenced its resonance, how string tension changed pitch, and how various construction techniques could produce different tonal qualities. This knowledge was not written down in physics textbooks or acoustics manuals. It was developed through centuries of experimentation, passed down through generations of craftsmen, and refined through constant practice and observation. By studying these instruments, we can begin to recover the scientific thinking of people who never wrote a single page of theory but understood the laws of physics better than many scholars of their time.

The challenge for historians is to develop ways of studying this practical science that do justice to its nature. We cannot simply treat it as a primitive version of modern science, waiting to be "discovered" and written down in proper scientific form. Nor can we romanticise it as pure intuition or cultural tradition. Instead, we need to understand it on its own terms. This requires a shift in how we do history. Instead of just reading manuscripts, we must learn to "read" the objects themselves, the instruments, the textiles, and the tools. We must listen to the oral traditions and study the apprenticeship (*gharana*) systems that kept this knowledge alive for a thousand years.[10]

In conclusion, the "historiographical problem" is not that medieval India lacked science; it is that our modern definitions of science are too narrow to see it. By expanding our view to include the "Invisible Laboratory" of the artisan and the "Practical Science”, both of which were sustained and transmitted through oral traditions, we can arrive at a more honest and complete understanding of Indian intellectual life. These knowledge systems did not rely primarily on written texts but were preserved, refined, and passed on through memory, apprenticeship, and performance. Seen in this light, we encounter a society in which the roles of botanist, physicist, and artist often converged in the same individual, and where some of the most sophisticated forms of scientific experimentation were taking place not in formal institutions, but within local workshops and lived practices.

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[9] Agrawal, Shyam S. “Excellence of Acoustics in India- Its Heritage and Recent Research.” *Current Science*, vol. 95, no. 5, 2008, pp. 641–648.

[10] Mishra, Arun “Bhartiya Kanth Sangeet aur Vadya Sangeet.” Kanishka Publishers, New Delhi, 2008, pp. 126-129.

Defining the Field: Science vs. Technology in the Medieval Workshop

To analyse musical instrument-making as a scientific field, we must move beyond the modern image of the scientist in a laboratory wearing a white coat and instead step into the dusty, vibrant world of the medieval workshop. Recovering this "Invisible Laboratory" requires us to examine what actually took place within its walls. In doing so, it becomes important to distinguish between **science** (the underlying principles of sound and material) and **technology** (the application and design of the instrument itself). In the context of the medieval Indian workshop, however, this distinction was not rigid. Science and technology were not separate domains but deeply interconnected practices, brought together through the embodied skills and experiences of the artisan. They were the two sides of the same coin, practiced by the same hands.

When we think of science in the medieval period, we often imagine astronomers observing the stars or mathematicians solving equations. However, the artisan working with wood, gut, and metal was engaged in a form of inquiry that was just as rigorous. This can be understood as a form of *lived science*, in which the craftsman functioned simultaneously as theorist and experimentalist. For the instrument maker, the science was not abstract knowledge but the underlying logic that guided every decision. It was the *why* and *what* behind every choice they made. When an artisan selected a specific piece of wood or adjusted the thickness of a resonator, they were not merely following a fixed procedure; they were applying a refined understanding of how materials, shapes, and tensions affected tonal quality, pitch, and durability. This constituted a form of practical science grounded in everyday experience. This understanding of acoustics-the science of sound, was not theoretical but embedded in practice, continuously tested and refined through repeated engagement in the workshop.

If science can be understood as the domain of underlying principles- *why* and the *what*, then technology represents the *how* of their application. The musical instrument itself, whether a complex veena with its large resonators or a simple flute, embodied forms of pre-colonial indigenous technology that were by no means inferior to large-scale state-sponsored developments such as irrigation or metallurgy.[11] Within the workshop, technology took shape as the practical application of scientific knowledge to address material and structural challenges. Every component of a medieval instrument functioned as a technological solution to a specific problem. For example, how do you keep a stringed instrument from warping under the immense tension of its strings? The solution was a technological one: the clever use of cross-bracing, the selection of aged wood that had lost its internal moisture, and the application of specific oils and resins to stabilise the material. Every curve of a sitar's neck and every placement of a fret was a technological choice designed to maximise the scientific potential of the materials.[12] Similarly, the use of large gourd resonators helped increase sound and control its quality.[13] In this way, artisans created sound technologies that were well suited to the cultural and environmental conditions of medieval India.This indigenous technology was perfectly adapted to the local environment. Unlike modern mass-produced instruments, medieval instruments were often custom-made to suit the specific humidity and temperature of their region. Techniques for stabilising

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[11] Chattopadhyay, Budhaditya. "On Pre-colonial Indigenous (Sound) Technologies." Proceedings of Politics of the Machines. *Rogue Research.* 2021, 14–17 Sept. 2021, p. 32.

[12] Sharma, Bhagwatsharan. "Sitar Malika." VI Edition, *Sangeet Press, Hathras (U.P.),* 1983, p. 14.

[13] Le Conte, Sandie, et al. "Acoustical Measurement of Indian Musical Instruments (Vina-s): Towards Greater Understanding for Better Conservation." *The Journal of the Acoustical Society of America*, vol. 123, no. 5, 2008, p. 3380.

wood, preventing warping, and maintaining string tension reflect a careful response to the challenges posed by the Indian climate.[14] In this sense, instrument-making was not merely a craft but a highly developed form of sound technology that demonstrates significant technical skill and understanding.

The selection of materials offers one of the clearest examples of the scientific knowledge present in the medieval workshop. What may appear to be a simple act of choosing wood was, in reality, a careful and informed process. Artisans did not select materials randomly or based only on appearance; they understood how different types of wood affected sound. However, research into the acoustical properties of wood shows that this selection was a highly sophisticated scientific process.[15] Medieval craftsmen had an empirical understanding of concepts like density and the sound radiation coefficient- the efficiency with which a material converts vibrations into audible sound waves. They knew that certain woods, like the Jackfruit wood (*Artocarpus heterophyllus*) often used for instruments such as the mridangam, or red sandalwood and sissoo for stringed instruments, were preferred because they provided the right balance between strength and lightness. Wood that was too heavy would reduce sound quality, while wood that was too weak could not withstand tension.[16]

way of "measuring" these properties was based on the senses, but it was careful and systematic. By tapping a piece of wood and listening to its sound, the craftsman measures the internal damping of the fibers, something we would today measure using modern instruments- a frequency analyser. By feeling the weight and grain of the wood, they were judging its density and stiffness. This knowledge was acquired through meticulous observation over many years of practice.[17] It represents a form of vernacular science, where materials were understood through direct interaction rather than written theory.

At the same time, this understanding was inseparable from technology as execution. Knowing that a certain type of wood produces a desirable sound was only one part of the process; the artisan also needed to shape that wood into a form that could effectively harness and project that sound. This brings us to the design of the resonator- the hollow body of the instrument that amplifies vibration. The shaping of a resonator is a clear example of technological skill grounded in scientific understanding. The size, curvature, and thickness of the resonating chamber determine how sound waves are reflected, absorbed, and amplified within it. A slight variation in shape can significantly alter the tonal quality of the instrument. In instruments such as the Ektara, the use of a dried gourd as a resonator demonstrates a sophisticated engagement with natural forms. The artisan selects a gourd of appropriate size and symmetry, hollows it out, and attaches it to a neck made of bamboo or wood. The resulting structure is not merely a container for sound but an active participant in shaping it. Here again, the distinction between science and technology becomes blurred. The artisan's ability to shape the resonator reflects technological expertise, but this expertise is guided by an understanding of how sound behaves within enclosed spaces.The workshop thus functioned as a space where knowledge was constantly tested and improved through practice.

A clear example of how science and practice worked together can be seen in how string tension affects pitch in instruments like the Ektara. The pitch of the sound produced depends on several factors: the length of the string, its thickness, and, crucially, the tension applied to it. Increasing the tension raises the pitch, while decreasing it lowers the pitch. This relationship, which can be expressed mathematically in modern physics, was understood by artisans through direct manipulation. They adjusted the tension by tightening or loosening the string using pegs or knots. Over time, they learned how much tension was needed

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[14] Chattopadhyay, "Pre-colonial Indigenous Technologies', p. 33.

[15] Brémaud, Iris. "Acoustical Properties of Wood in String Instruments Soundboards and Tuned Idiophones: Biological and Cultural Diversity." *Journal of the Acoustical Society of America*, vol. 131, 2012, pp. 807–818.

[16] Ibid., p. 811

[17] Ibid., p. 814.

to produce the right sound without breaking the string.This knowledge was not fixed but developed through constant practice. If a string broke, the artisan adjusted their method. If the sound was not right, they changed the material or structure. In this way, learning happened through repeated trial and correction. Even without formal tools or theory, this process closely resembled experimentation, taking place within the everyday space of the workshop.

What emerges from these examples is a picture of the medieval workshop as a space where science and technology are not sequential but simultaneous. The artisan does not move from theory to application; rather, theory is embedded within practice. Each act of making was also an act of knowing, and every finished instrument reflected set of principles that had been tested and validated through experience. By recognising the distinction between the *science* of principles and the *technology* of application, we can better understand the role of artisans in the history of science. They were engineers who developed effective sound technologies, many of which are still valued today for their quality and craftsmanship.

The Invisible Laboratory as a Way of Knowing

The idea of the "Invisible Laboratory," introduced at the beginning of this paper, can now be revisited with greater clarity and depth. It is not merely a metaphor used to describe the absence of formal scientific institutions in medieval India; rather, it points to a fundamentally different mode of producing, organising, and validating knowledge. If modern science is often associated with the controlled, visible space of the laboratory, equipped with instruments, measurements, and written records, the medieval workshop functioned as a dispersed and embodied counterpart. It was invisible not because it lacked rigour, but because its processes were not codified in forms that modern historiography readily recognises.

In the context of musical instrument making, this invisible laboratory was not confined to a single physical location. It extended across multiple spaces: the artisan's workshop, the site of performance, and the everyday environments in which instruments were used and maintained. Knowledge did not reside in a fixed archive but circulated through these spaces, constantly tested and refined through practice. The making of an instrument, its tuning, its performance, and even its repair were all part of a continuous process of experimentation. Each stage provided feedback, allowing the artisan to adjust materials, techniques, and designs in response to the desired acoustic outcome.

What distinguishes this mode of knowledge production is its deeply embodied character. Unlike textual knowledge, which is externalised and preserved in written form, the knowledge of the instrument maker was internalised within the body, within the trained hands that shaped the material, the ear that judged tonal quality, and the memory that retained techniques across generations. This embodied knowledge cannot be easily separated from the individual who possesses it. It is acquired through long periods of apprenticeship, where learning takes place not through abstract instruction but through observation, imitation, and repetition. The novice watches the master, listens carefully, and gradually develops the sensory and motor skills required to replicate the process.

In such a system, the hand and the ear are the main tools for learning. The artisan's hand learns to recognise the resistance of wood, the flexibility of bamboo, or the tension of a string. At the same time, the ear becomes attuned to subtle variations in sound, such as pitch, resonance, and tone, which show whether the instrument is made properly. This sensory engagement replaces the need for written formulas or numerical measurements. Instead of calculating frequency or amplitude, the artisan evaluates sound directly, using skills that are precise even without words.

This leads to an important idea: in the "Invisible Laboratory," sound itself becomes a form of knowledge. It is not merely the end product of the instrument but also the means through which the instrument is tested and validated. When an artisan makes or repairs an instrument like the Ektara, the final judgment is based on how it sounds, not just how it looks. If the sound is clear and steady, the work is successful. If it sounds dull or unstable, changes are made. In this sense, sound functions as both evidence and verification, a direct indicator of whether the underlying principles have been correctly applied.

Understanding the Invisible Laboratory in this way allows us to rethink the role of the artisan. Far from being a mere craftsman executing predefined tasks, the instrument maker emerges as an experimenter and innovator. Each act of making involves a series of decisions about material selection, structural design, and acoustic adjustment, that are informed by prior experience but also open to modification. When a particular technique produces an improved sound, it is retained and transmitted; when it fails, it is revised or discarded. This process of trial and improvement resembles what we might describe as experimentation, though it is conducted without formal protocols or written records.

Importantly, this knowledge is not purely individual. It is embedded within communities of practice, families, guilds, and performance traditions, where it is collectively maintained and transmitted. The "Invisible Laboratory" was therefore both personal and social. It existed in the skills of the artisan, but also in the shared practices and memories of the community. This dual nature ensures both continuity and adaptability: continuity, because techniques are preserved across generations; adaptability, because each practitioner can modify them in response to changing materials, environments, or aesthetic preferences.

By conceptualising the medieval workshop as an Invisible Laboratory, we move beyond the limitations of text-centric historiography and recognise the richness of practice-based knowledge systems. Science, in this context, is not confined to theoretical abstraction but is embedded in material engagement and sensory experience. Technology was not only about large machines but also about carefully made objects that used natural principles in precise ways. Together, they formed a way of knowing based on doing, listening, and remembering.

As this section shows, the idea of the "Invisible Laboratory" helps us understand science and technology in a more grounded and inclusive way. It shows that systematic knowledge can exist even without written texts, and that the boundaries between science, technology, and art are often not fixed. In musical instrument making, these boundaries come together, creating a form of knowledge that is scientific, practical, and cultural at the same time.

To pursue this insight further, however, we now need to ask how this knowledge was passed on. If it was not written down, how did it continue across generations? The next section looks at this question by focusing on oral traditions, apprenticeship, and systems of learning based on practice.

Oral Traditions as Scientific Repositories

In the historiography of science and technology, written texts have often been treated as the main source of knowledge. Because of this, societies that relied on oral traditions are sometimes seen as "pre-scientific" or "mythological." However, this assumption does not hold true in the context of medieval India. A rich and complex body of knowledge existed outside written texts, especially in oral traditions. This chapter argues that such oral traditions, particularly those related to musical instrument making, functioned as important repositories of knowledge. In this sense, what we may call the "oral knowledge system" becomes central. Here, knowledge was not stored in books but practiced, remembered, and performed.

The concept of the "oral knowledge system" challenges the traditional archival paradigm that equates "truth" with textual documentation. In medieval India, memory and practice played a key role in preserving knowledge about acoustics, materials, and construction techniques.[18] This archive is not merely a collection of sounds but a cognitive system where the ear is trained to detect microscopic variations in pitch, timbre, and vibration; variations that no medieval text could sufficiently quantify or describe.[19] In this system, the artisan's ear became the main tool for judging accuracy, while the apprentice's memory became the place where this knowledge was stored and passed on. This is clearly seen in instruments like the *Ektara* and *Ravanhatta*, where making the instrument required a strong practical understanding of materials and sound, even without written instructions.The construction of these instruments required a deep, albeit non-textual, understanding of physical laws.

To understand how scientific knowledge was transmitted orally, it is important to look at it through the framework of memory studies. In pre-modern societies, knowledge did not exist only in written texts but was also preserved through memory and oral traditions. As scholars like Barbara A. Misztal in *Theories of Social Remembering* discusses that in oral cultures, the past is closely tied to the present, and knowledge is maintained through shared practices, habits, and collective understanding rather than through fixed written records.[20] In the context of medieval Indian workshops, memory functioned as a collective resource, held and validated by communities of artisans. Knowledge about instrument-making : how to shape a resonator or adjust string tension, was not written down but remembered, practiced, and corrected within the group. This also means that memory was not a perfect record of the past, but a "selective reconstruction," shaped by present needs and practical use.[21] As a result, only the knowledge that remained relevant for making and playing instruments was preserved over time, while less useful details were gradually forgotten.

Barbara Misztal's work shows that in oral societies, people do not simply "store" knowledge; they actively shape and adapt it. The past is remembered only if it remains useful in the present. This dynamism is central to our argument. Knowledge that is no longer useful is gradually forgotten, a process often described as "structural amnesia."[22] In this way, memory is not passive but active, keeping only what serves a purpose. Within artisan communities, this shared memory allowed new techniques and ideas to be absorbed into the ongoing tradition of the craft.

The difference between history and memory is often understood as the difference between written and oral knowledge. History aims to produce an objective record of the past, while memory exists through the lived practices of communities. Despite the invention of writing, antiquity was still dominated by oral memory, often blending myth and history without clearly separating them. Memory was highly valued across traditions. This becomes especially important in technical crafts like instrument-making, where knowledge was rarely written down but was continuously practiced, remembered, and passed on.

In the context of medieval India, the lack of written manuals on instrument making has often been misunderstood as a lack of proper or systematic knowledge. However, this is not true. As scholars argue, the absence of texts does not mean the absence of rigour. Knowledge was mainly passed down orally, often within families. In instruments like the Ektara and Ravanhatta, skills such as shaping, tuning, and assembling were preserved through memory and repeated practice rather than written instructions.

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[18] Brita, Antonella, et al., editors. Manuscripts and Performances in Religions, Arts, and Sciences. *Boston*, 2024. p.

[19] Sayers, Emily. "Pedagogical Strategies for the Development of Improvisation and Composition in North Indian Classical Music." *Frontiers in Psychology*, vol. 16, 2025.

[20] Lowenthel cited in Misztal, Barbara A. *Theories of Social Remembering*. Maidenhead: Open University Press, 2003, pp. 28-29.

[21] Ibid., 29

[22] Douglas cited in Ibid., 30

Even though medieval society was not highly literate in the modern sense, it was still very aware of writing and its uses. Until the late Middle Ages, writing was considered a support to memory rather than a replacement for it. Knowledge was valued as something to be remembered and practiced. The past was understood in relation to present needs, which kept techniques flexible and changing. This explains why instrument-making knowledge remained flexible and changing, artisans modified techniques based on present musical needs rather than preserving a fixed ‘original' method. The choice to rely on oral transmission in certain fields, especially technical crafts, was often deliberate. Knowledge was not freely shared; it was carefully protected within artisan groups, families, or caste-based communities and reserve it for their descendants. For instrument makers, this also functioned as a form of control over specialised knowledge- ensuring that techniques of sound production, material treatment, and design remained within specific lineages.

With the rise of printing and the spread of knowledge through books, mnemonic techniques began to fade.[23] By the seventeenth century, scholars increasingly moved away from memory-based systems, and started relying more on written texts, and knowledge became something to be recorded and studied rather than simply remembered. Society began to appreciate the "pastness" of the past.[24] Renaissance scholars focused on reading texts carefully and checking their authenticity. As a result, knowledge that was not written down, such as craft knowledge, started to be seen as less reliable or less “scientific.” The growth of archives, printed books, and new ways of recording information strengthened this shift.[25] Over time, written documents became more important, and knowledge was increasingly organised through texts. In this process, the practical and embodied knowledge of artisans such as how to control vibration or resonance, was increasingly pushed to the margins of what counted as legitimate knowledge.

This change was also linked to a new way of thinking about time. Instead of remembering through lived experience and shared practices, knowledge began to be organised in chronological order. The eighteenth century was particularly significant because the meaning of memory both expanded and weakened. First, the rise in literacy and the expansion of scientific knowledge stretched memory's capacity, but the vast growth of specialised disciplines and books meant that even a well-trained memory could no longer hold all knowledge. This led to the creation of dictionaries, encyclopaedias, museums, archives, and libraries to store and organise knowledge. At the same time, knowledge of instrument-making, that depended on practice could not easily be transferred into these new textual and classificatory systems.

The Enlightenment further reduced the importance of memory by placing greater value on reason in the arts and sciences.[26] Thinkers of this period began to reject tradition, often associating it with backwardness. As a result, practices rooted in oral transmission, such as musical instrument making, were overlooked or undervalued. The knowledge involved in making instruments like the Ektara and Ravanhatta was no longer seen as scientific, but merely traditional or artisanal. Yet, this judgment ignored the fact that these instruments required precise control over sound, material, and structure that demonstrates a form of practical science developed outside written theory. Historians influenced by Enlightenment thinking began to judge the past through the lens of rationality and progress, often portraying earlier knowledge systems as irrational or undeveloped. In this process, the complex material understanding and acoustic knowledge involved in instrument making were ignored, simply because they were not expressed in written or formal scientific terms.

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[23] Ibid., 36

[24] Douglas ,35

[25] Ibid., 36

[26] Ibid., 37-38

By the late nineteenth and early twentieth centuries, a 'crisis of memory' emerged as modernisation and industrialisation disrupted traditional ways of life. As societies became more focused on scientific progress and rationality, many oral traditions, including those related to instrument making, began to decline. The fear that such knowledge might disappear led to efforts to preserve it, sometimes by documenting instruments or standardising their forms. However, much of the original, practice-based knowledge remained difficult to capture in written form.

In the early twentieth century, memory itself became an important area of study, especially through thinkers like Maurice Halbwachs, who argued that memory is shaped by social groups, not just individuals.[27] This is especially relevant for instrument-making traditions, where knowledge survives only as long as the community continues to practice it. As these communities change or weaken, this shared knowledge also becomes fragile and may gradually disappear or transform. In short, memory, which once formed the foundation of knowledge systems, becomes more complex and contested over time. In the context of medieval India, this insight is important for understanding how oral traditions functioned as repositories of technical knowledge. The knowledge embedded in practices such as the making of instruments like the *Ektara* and *Ravanhatta* was never fixed or static; rather, it was continuously reshaped through performance, practice, and transmission within specific communities. Each act of making or playing an instrument was also a way of remembering and passing on knowledge.

At the same time, the weakening of strong, shared memory systems, especially under processes of social and technological change highlights the fragility of such knowledge traditions.[28] When collective memory declines, the continuity of orally transmitted technical knowledge becomes uncertain. This helps explain why many forms of artisanal and musical knowledge today appear fragmented or are no longer recognised as systematic or scientific. However, such knowledge does not disappear entirely; it survives within smaller, community-based traditions, where memory continues to be preserved and practiced. In these settings, instruments are not just used to produce sound but also express cultural identity. Oral traditions, therefore, should be seen as active systems that continue to preserve and pass on knowledge across generations.[29]

Oral knowledge systems were equally sophisticated but differently structured than textual ones. While a text is vulnerable to physical decay or the loss of its linguistic key, oral knowledge survives through continuous practice and performance. For both scholars and artisans in the medieval period, knowledge was often shared through oral performance rather than written texts. Whether it was mathematics, astronomy, or technical skills, learning took place through speaking, listening, and demonstration. In many cases, knowledge was preserved more in people's minds than in written texts. This creates what may be called "paradox of sources," where important technical knowledge existed but was intentionally kept outside written records to maintain control within specific groups.

In the absence of detailed written manuals, knowledge existed in practice rather than text. What may be termed *savoir-faire*[30] that is practical, non-literary knowledge which has formed the basis of technological understanding. The artisan's workshop functioned as a site of continuous knowledge production, where techniques were transmitted through demonstration, imitation, and repetition.

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[27] Ibid., 45-46

[28] Misztal, Theories of Social Remembering, 58

[29] Barbara A. Misztal, *Theories of Social Remembering* (Maidenhead: Open University Press, 2003), pp. 28–30.

[30] For the concept of *savoir-faire* as embodied technical knowledge, see Bray, Francesca, "Technics and Civilization in Late Imperial China," *Science and Civilisation in China*; and Eyferth, Jacob, *Eating Rice from Bamboo Roots*. Harvard University Press, 2010.

Building on the idea that knowledge was preserved within smaller, community-based traditions, it is important to see how this knowledge did not remain confined to a single place. Instead, it moved through people, performances, and travelling traditions. Across medieval India, there existed groups of wandering performers, mendicants, and musician communities who carried not just music, but knowledge across regions. Their movement turned oral tradition into a mobile system of transmission.[31] Here, storytelling, performance, and musical practice become part of a shared knowledge network. These performers were not merely entertainers but were carriers of knowledge. Through repeated performance, they transmitted stories, ethical ideas, and local world-views, while also sharing practical understandings of geography, ecology, and everyday life. Instruments travelled with them and became tools through which knowledge moved across space.

One of the clearest examples of this is the Baul tradition of Bengal, closely associated with the Ektara. For the Bauls, music was never just entertainment. As they themselves understand it, *gaan* (song) carries a double meaning, it is both music and knowledge.[32] This idea is crucial for our argument. When Bauls sing, they are not simply performing; they are teaching, sharing, and transmitting ways of understanding the world. Their songs often contain reflections on the body, nature, spirituality, and everyday life, all expressed in simple yet layered language. In this sense, the Ektara becomes more than an instrument, it becomes a medium through which knowledge travels.

Similarly, Rajasthani bardic traditions associated with the *Ravanhatta*. In Rajasthan, itinerant musicians and bards moved from one region to another, performing in villages, courts, and local gatherings. The Ravanhatta, played by the bards and minstrels of the Nayaka community in Rajasthan and Gujarat. Known as *Bhopas*, these minstrels perform religious-themed songs in honour of Pabuji, the folk deity of the Rabari or shepherd community of Rajasthan.[33] Hundreds and thousands of these wandering minstrels are earning their livelihood by narrating the epics while playing the instrument. Through their performances, they carried stories of kings, genealogies, local heroes, and moral histories. But along these narratives, they also transmitted practical knowledge- about routes, deserts, seasonal changes, and survival practices. Their movement across spaces allowed for an exchange of ideas, where knowledge was not fixed but constantly reshaped through interaction with different environments and audiences. Without any formal education, these musicians learn just by listening, and have had been able to preserve the traditional repertoire, intact.

What connects these different traditions is the idea that knowledge was not stored in one place but moved with people. Oral traditions made this movement possible. Through songs, performances, and repeated practice, knowledge about regions- their geography, climate, available materials, and techniques, was shared across boundaries. A performer travelling from one region to another did not just bring music; they brought with them an entire system of understanding shaped by their local environment. A song could teach values, narrate history, and encode environmental knowledge simultaneously. Knowledge was not separated into categories such as “science,” “history,” or “art.” Instead, it existed as an integrated experience transmitted through performance.

Seen from this perspective, instruments like the Ektara and *Ravanhatta* are not just musical objects. They are mobile knowledge systems. They allow us to understand how medieval and folk societies preserved and transmitted complex forms of understanding without relying on written manuals or formal documentation. The performer becomes a carrier of memory, and the performance becomes an archive in motion. The Baul idea of *gaan* captures this perfectly, music and knowledge are not separate but deeply

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[31] Burke, Peter. ‘History as Social Memory’, in *Varieties of Cultural History*, Cornell University Press, New York, 1997, 44-45.

[32] Openshaw, Jeanne. *Seeking Bauls of Bengal*. Cambridge University Press, 2002, p. 88.

[33]Kothari, Komal. “Performing Pabuji: Narrative Traditions of Rajasthan.” *Oral Epics of India*, Rajasthan Sangeet Natak Akademi; see also Harlan, Lindsey. *Religion and Rajput Women*.

interconnected. To sing was to know, and to perform was to transmit that knowledge further. Thus, oral traditions challenge the idea that knowledge must be fixed in texts to be reliable. Instead, they show that knowledge can survive through repetition, movement, and performance. Oral tradition, therefore, was not a secondary system of memory, rather it was a primary mode of intellectual and cultural transmission.

Alongside oral transmission, *guru-shishya parampara* or *gharana* (apprenticeship) formed the core structure through which this knowledge was learned and sustained.[34] In medieval Indian craft and performance traditions, learning did not take place through formal instruction but through close, long-term engagement between master and apprentice. The apprentice observed the master at work, gradually imitated techniques, and refined their skills through repeated practice. This process involved not only mechanical learning but also the training of the senses: the hand learned to feel the texture and resistance of materials, while the ear became sensitive to subtle differences in sound, pitch, and resonance.

This system is clearly visible in traditions associated with instruments like the Ravanhatta and the Ektara. Among the Bhopa performers of Rajasthan, who narrate the epic of Pabuji using the Ravanhatta, knowledge is passed down within families or closely connected communities. Young apprentices learn by accompanying senior performers, observing both musical technique and narrative performance, and gradually participating in the act itself. Similarly, in the *Baul* tradition of Bengal, closely linked with the Ektara, learning takes place through lived experience and association with a guru. The disciple absorbs not only musical skills but also philosophical ideas, performance styles, and ways of interpreting the world through song.

In both cases, apprenticeship functions as a structured yet flexible system of learning, where knowledge is transmitted through embodied experience, discipline, and participation in a community of practice rather than through written rules. This ensures continuity while also allowing for adaptation, as each generation learns, modifies, and carries the tradition forward.

Material Science and Ecological Selection:

The discussion of oral traditions as repositories of knowledge becomes particularly significant when we examine the kind of knowledge they preserved. Among the most important forms of such knowledge was an intimate understanding of materials and their relationship with the surrounding environment. This engagement was neither incidental nor purely utilitarian; rather, it constituted a systematic and experience-based mode of knowledge production. In the context of musical instrument making, this is especially evident in the careful selection, treatment, and manipulation of natural resources, reflecting what may be described as an embodied and ecologically grounded material science.[35] Unlike the abstract, text-heavy scientific traditions of the period, instrument making was a "field science" rooted in the critical evaluation of organic materials. In this setting, the making of musical instruments can be understood as a form of practical material science, one that was not codified in written texts but deeply embedded in artisanal practices and transmitted through oral and experiential means.

At its core, material selection in instrument making was a deliberate and analytical process. This selection process was fundamentally ecological. Craftspeople did not merely rely on available resources; rather, they assessed materials such as wood, animal hide, plant fibres, and metals for their density, flexibility, tensile strength, and resonance. These properties directly influenced the tonal quality, durability, and functionality of the instrument. The geography of India, ranging from the humid deltas of Bengal to the arid plains of Rajasthan, dictated a specific material logic. For instance, the use of seasoned gourds (*Lagenaria siceraria*) in the construction of stringed instruments was not an arbitrary choice but a scientific

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[34] Mishra, Kanth Sangeet aur Vadya Sangeet, p. 126-27.

[35] Dawe, Kevin. Musical Instruments as Material Culture. Routledge, 2023, p. 89.

optimisation based on the material's lightweight nature and high internal volume, which allowed for significant sound amplification without adding excessive weight.[36] Artisans tested these gourds by gently tapping them and listening to the sound they produced. This helped them judge whether the thickness was even and whether there were any cracks inside.

Furthermore, this ecological sensitivity is particularly evident in practices surrounding wood selection. The choice of wood depended not only on availability but also on its response to climatic conditions such as humidity and temperature. In regions with high moisture, certain woods were avoided due to their susceptibility to warping, while in drier climates, denser woods were preferred for their stability and acoustic depth. Artisans chose woods like jackfruit (Artocarpus heterophyllus) and teak (Tectona grandis) based on how well they produced and carried sound.[37] Craftspeople preferred trees that were at least thirty years old, ensuring sufficient maturity and density. Trees growing near water sources were avoided, as excessive moisture weakened the wood, reducing its structural integrity and acoustic potential. Similarly, logs containing knots (suli) were rejected because they compromised the strength of the instrument body. The timing of tree felling was also critical: artisans waited until the tree had shed its leaves, indicating a particular stage in its biological cycle, before cutting it down. Once harvested, the wood was carefully seasoned in open spaces to reduce moisture content and enhance durability. These practices underscore a systematic engagement with ecological processes, where knowledge of plant growth, climate, and material transformation informed each stage of production. Thus, we can say that Artisans were not merely passive users of the materials provided by their local environment; they were active researchers who evaluated material properties. Such decisions reveal an empirical understanding of material behaviour that aligns closely with principles of modern material science, albeit articulated outside formal scientific vocabulary.

This ecological sensitivity extended further into the selection and preparation of structural materials for instruments such as the *Ektara* and the R*avanhatta*, where craftsmen demonstrated a careful and context-specific understanding of plant and organic resources. While the *Ektara*, widely used in Bengal and eastern regions, typically employed a dried bottle gourd (tumba or lau) or sometimes coconut shell as its resonator, the *Ravanhatta* of Rajasthan relied instead on a coconut shell. This difference was not merely stylistic but deeply ecological. In the riverine and fertile landscapes of Bengal, gourds were abundant, lightweight, and acoustically effective, making them an ideal choice for itinerant musicians who required portability alongside resonance. They used bamboo for the neck as they are flexible. This flexibility is crucial because, in many forms of the *Ektara*, the player actively compresses or bends the bamboo neck to change string tension and produce pitch variations. A rigid material like hardwood would not allow this kind of dynamic modulation. So the choice of bamboo directly shapes the musical technique itself, not just the structure. There's also a climatic dimension. In humid environments like Bengal, bamboo performs better than many woods because it is less prone to cracking or warping when properly cured. This makes it more durable under fluctuating moisture conditions.

By contrast, in the more arid and semi-arid regions of Rajasthan, where such gourds were less readily available, artisans adapted by using coconut shells or occasionally carved wooden bowls as the resonator. Its hemispherical shape helps trap and reflect sound waves, amplifying the vibrations produced by the strings. The open side of the coconut shell is covered with goat skin, which plays a very important role in sound production. The goat skin is stretched tightly over the shell to form a thin surface that vibrates when the strings are played.Goat skin is used because it is strong but also flexible enough to stretch without breaking. The thickness of the skin also affects the sound, thinner skin produces sharper sounds, while thicker skin gives a deeper tone. This shows that instrument makers understood how different materials

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[36] Conte, Davide, et al. "Acoustical Measurement of Indian Musical Instruments (Vina-s): Towards Greater Understanding for Better Conservation." *The Journal of the Acoustical Society of America*, 2008, p. 5641.

[37] Brémaud, Iris. "Acoustical Properties of Wood in String Instruments Soundboards and Tuned Idiophones: Biological and Cultural Diversity." *The Journal of the Acoustical Society of America*, 2012, p. 816.

affect sound. The long neck of the *Ravanhatta* is usually made from bamboo. Bamboo is chosen because it is light, strong, and slightly flexible. Made from locally available material, the instrument is very easy to make, and generally, *Ravanhatta* players make their own instrument. Though, these musicians do not know any vocabulary and terminology of music, yet they tune their *Ravanhatta* perfectly and sing in unison with their instrument. The bow (*Gaj*) used for playing the *Ravanhatta* is typically made from lightweight, locally available wood. In Rajasthan and nearby areas, craftsmen commonly use woods such as bamboo, neem (Margosa), or other flexible local hardwoods.

Importantly, this knowledge did not reside solely with artisans but was distributed across a network of cultivators, craftsmen, and musicians. Plantation owners cultivated specific varieties of jackfruit trees not for fruit yield but for high-quality timber, employing techniques such as controlled irrigation and early harvesting to optimise wood strength. Instrument makers, in turn, developed relationships of trust with these cultivators, often inspecting trees personally before purchase. Musicians also played an active role in material selection, as the sound quality of the instrument directly affected their performance. The collaborative selection of a wood log, sometimes described metaphorically as a form of "wedlock," illustrates how knowledge circulated across different actors, forming a shared epistemic community. In this sense, material knowledge was not individual but collective and relational, emerging from continuous interaction between humans and their environment.

If wood, gourds, and membranes formed the structural body of instruments such as the *ektara* and *Ravanhatta*, the strings can be seen as their soul, which constituted their acoustic core, translating material knowledge into sound. The making of strings in the medieval period reflects an equally sophisticated understanding where flexibility, friction, and durability, were grounded in locally available resources and refined through artisanal practice.

In the case of the *Ektara*, strings were traditionally made from natural gut (animal intestines), plant fibres, or later metal wire, each selected for its specific acoustic and mechanical properties. Gut strings, derived from animal intestines, were valued for their elasticity and warm tonal quality, but required careful preparation. The intestines were cleaned, stretched, twisted, and dried under controlled conditions to achieve the desired thickness and strength. This process demanded precision: excessive drying could make the string brittle, while insufficient curing would reduce its tensile strength and stability. In some regions, plant-based fibres were twisted into strings, offering a more readily available but less durable alternative. Over time, the adoption of metal strings, particularly iron or steel, reflected an evolving material adaptation, as these provided greater durability and louder projection, especially suited to outdoor performance contexts.

For the *Ravanhatta*, the use of horsehair strings represents a distinctive and highly specialised material choice. Horsehair, obtained from the tail, was prized for its combination of strength, flexibility, and responsiveness to friction, making it particularly suitable for bowed instruments. Artisans carefully selected and bundled strands of hair, ensuring uniform thickness and tension. Interestingly, the hair are coated with a powder cake (Bejara)[38],which improved friction between the bow and the strings. This treatment reflects an implicit understanding of frictional dynamics, allowing for controlled vibration and consistent tonal output during performance.[39]

The structural components of the instrument also reveal a thoughtful engagement with material properties. The wood used, often sourced from species like teak, rosewood (such as Indian rosewood or sheesham), or mahogany, were chosen for their durability and ability to support resonance and tonal depth. These woods were sometimes treated with natural varnishes derived from resins and oils, which not only

[38] https://gaatha.com/the-story-of-a-storyteller-his-instrument/

[39] Aggarwal, Neha, and Chander Pal Punia. "Chemical Analysis and Conservation of Materials in Indian Musical Instruments." Vol. 14, No. 1, *Sangeet Galaxy e-Journal, Haryana. 2005,* pp. 28-31.

enhanced their visual appearance but also protected them from environmental damage such as moisture and cracking. In addition, traditional decorative practices, including the use of natural dyes from plant and mineral sources, contributed both to the aesthetic value of the instrument and to its preservation by forming a protective layer against climatic exposure.

Durability remained a central concern in the preparation and maintenance of strings across both the *Ektara* and the *Ravanhatta*. Musicians developed practical techniques to protect strings from environmental stress, particularly fluctuations in humidity and temperature, which could weaken organic materials. Strings were regularly cleaned, adjusted, and replaced when necessary, while tension was carefully modified according to climatic conditions. In the case of gut or plant-based strings, periodic oiling helped maintain flexibility and prevent cracking, whereas horsehair strings required careful handling to avoid fraying. This body of knowledge, transmitted through experience and practice rather than formal documentation, enabled artisans and musicians to maintain consistency in sound quality and performance. It highlights a deeply empirical understanding of material behaviour, where maintenance itself became an extension of the broader material science underlying instrument making.

Conclusion

This study set out to rethink how we understand science and technology in medieval India by moving beyond a narrow focus on written texts and elite knowledge systems. By examining musical instrument-making, it becomes clear that scientific knowledge did not exist only in books or royal courts, but was also deeply embedded in everyday practices of artisans and performer communities. Instruments such as the *Ektara* and the *Ravanhatta* demonstrate that processes like material selection, sound production, and structural design required careful observation, testing, and refinement- what can be understood as a form of practical science.

At the centre of this study has been the idea of oral traditions as key carriers of knowledge. Unlike written texts, which preserve knowledge in fixed form, oral systems kept knowledge alive through memory, practice, and performance. Artisans learned through apprenticeship, where skills were passed on by doing, listening, and repeating rather than reading. This meant that knowledge of acoustics, materials, and environmental adaptation was not absent, but simply existed in a different form- embodied, flexible, and constantly evolving. The concept of the "Invisible Laboratory" helps us understand this process more clearly. The workshop, performance space, and everyday environment together formed a space of continuous experimentation. Here, knowledge was tested through sound, corrected through practice, and preserved through community memory.The study also showed that material knowledge was closely connected to ecology. The selection of wood, gourds, skins, and strings depended on regional climate and available resources. This reflects a deep ecological awareness, where artisans adapted their techniques to suit different environments. Such practices demonstrate that medieval instrument-making was not random or purely traditional, but based on careful and experience-based understanding of materials.

Overall, this paper argues that the absence of written manuals does not mean the absence of science. Instead, it highlights the limitations of text-based historiography. By focusing on oral traditions, craft practices, and material engagement, we gain a more inclusive and accurate understanding of knowledge systems in medieval India. Musical instrument-making, therefore, emerges as an important site where science, technology, and culture come together- showing that knowledge was not only written, but also lived, practiced, and remembered.

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ANNEXURE

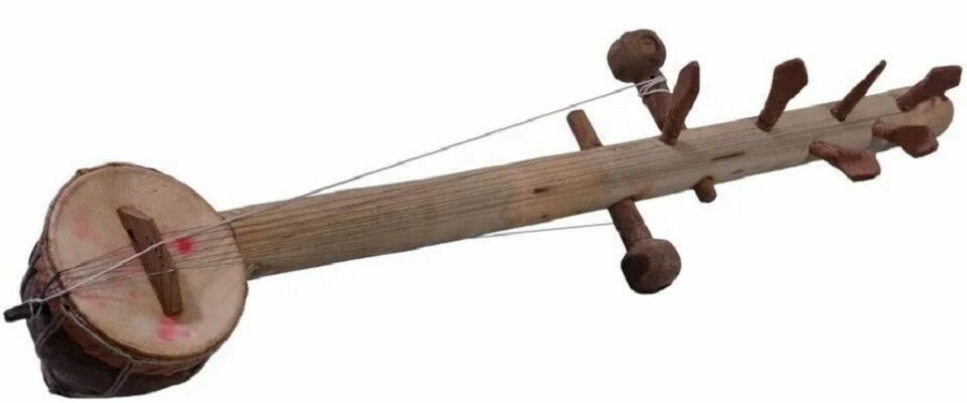


Fig. 1 : Ravanhatta- used by Bhopa Performers in Western Part of the subcontinent.
https://share.google/93SaUCGddkhHfnOVV

Fig. 2&3 : Raw materials used in the making of Ravanhatta
https://gaatha.com/the-story-of-a-storyteller-his-instrument/

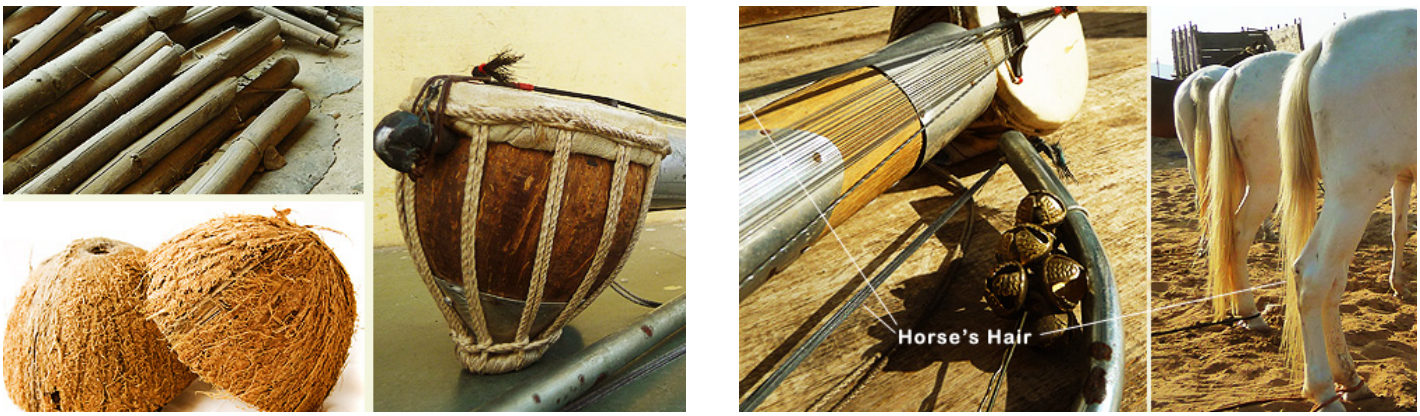




Fig 4 : Ektara used by the Baul Community in Bengal.
https://share.google/EWX4S7qLkE676w6Nj

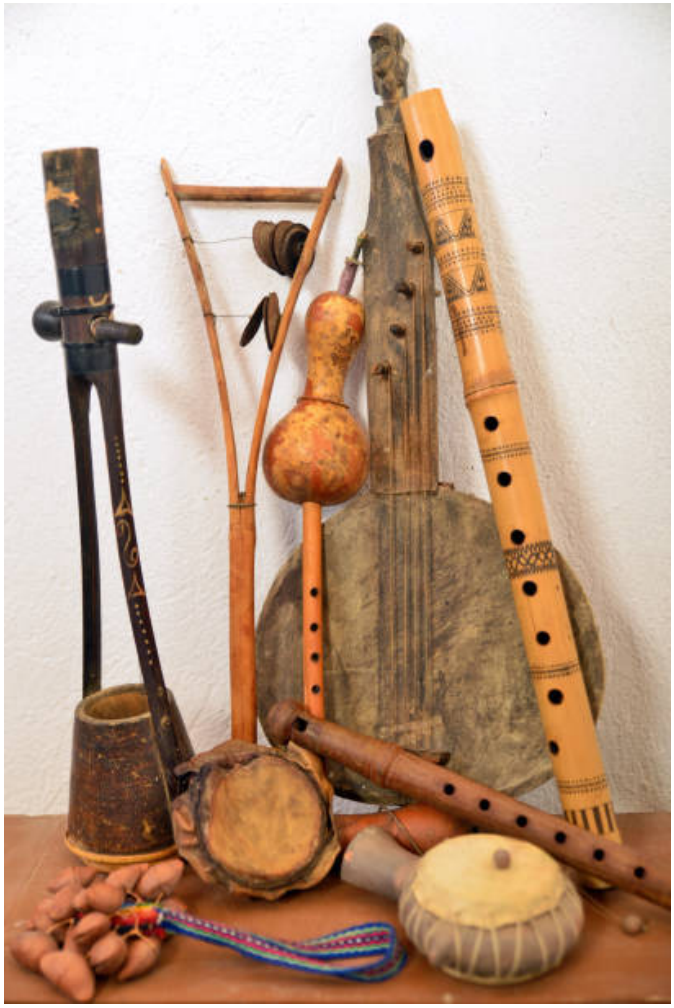


Fig. 5 : Ekatara in the Left corner
https://www.istockphoto.com/photos/ektara